

An 8 MHz Fully-Differential Switched-Capacitor Voltage Doubler for 8-bit ADCs

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Abstract

A 8 MHz fully-differential switched-capacitor (SC) voltage doubler circuit, to be interfaced with an 8-bit analog-to-digital converter (ADC), was implemented in TSMC 180 nm CMOS technology, operating from a 1.8V supply voltage. The amplifier provides differential outputs with an output swing of 1.4V peak-to-peak differential (0.7V per side). It contains a SC network (with CMOS sampling switches) and a fully-differential operational transconductance amplifier (OTA). In pipelined ADC architectures, the SC amplifier is one of the most critical building blocks, because any errors introduced are amplified and propagated through the converter, significantly affecting the overall system resolution and linearity.

Since the doubler outputs a differential swing of 1.4 V, the differential LSB would be approximately 11 mV. The power dissipation is 1.45 mW operating on a 1.8V supply in a 180nm process. The OTA is designed with 73.7 dB of open loop gain, with a 3 dB bandwidth of 123 kHz, gain crossover of 321 MHz, and phase margin of 48.5°. The slew rate is 4.6 V/ns.

Introduction

Fully differential amplifiers provide better common mode rejection than single-ended topologies, since they cancel the noise that typically originates from the common mode (CM). Switched capacitors offer miniaturization and tunability that resistive feedback cannot provide. There is also a reduced thermal noise in the circuit, increasing the accuracy.

Design Architecture

The fully-differential SC doubler follows a sample and hold, non-overlapping clock topology in which the capacitors maintain the total sample charge during the respective clock phase. The capacitors are referenced to the CM on either side and therefore symmetrical. The design is shown below in Figure 1.

The sample and hold clocks have a 0.5 ns deadtime to avoid switch shoot-through. During sampling, all capacitors are charged to the input voltage. Once the clock switches to the hold cycle, the Y-Z capacitors receive a zero voltage (since Z is tied to CM and Y becomes CM via negative feedback). Their charge must be transferred to the X-Y capacitors. The charge transfer results in double the input voltage across the X-Y capacitors, so this topology is inherently a double-gain circuit.

The fully-differential OTA functions as negative feedback control to bring both of its inputs ("Y" nodes) to the CM voltage. The outputs are fed into a common mode feedback resistor divider to continuously detect the output CM. A single-ended amplifier is used to reference the ideal CM (0.9 V) and output gate biases into the tail transistors of the differential OTA. These gate biases modulate the tail currents, in turn shifting the CM output in the correct direction.

The OTA was chosen to be a three-stage design to achieve the required gain of over 60dB. This topology allows for high gain, while maintaining high bandwidth. The folded-cascode OTA utilizes the cascode output to achieve the required gain, but this causes the devices to be larger. As the device size increases, the parasitic capacitors also increase, causing the bandwidth to decrease. Thus, a three-stage OTA was chosen due to the high speeds required.

Design considerations

For this design, specifications were selected to maintain accurate performance across the full scale voltage (V_{FS}). The output range for V_{FS} was chosen conservatively, to allow 200mV of headroom on either side of the common mode voltage (V_{CM}); with a supply voltage of 1.8, this results in a V_{FS} of 1.4V. V_{CM} was chosen as 900 mV (the center voltage), so signals would be centered between the supply terminals and maximize swing in either direction. Based on an 8-bit ADC specification, the LSB

voltage will be $2 \cdot V_{FS}/2^8 = 10.9\text{mV}$. In order to maintain accuracy down to 8 bits, the amplifier error must be no more than $V_{LSB}/2$, which is 5.5mV .

The system output noise was $V_{LSB}^2 = 120 \text{ mV}^2$. At room temperature $T=300 \text{ K}$, C_{samp} must be greater than 0.831 fF in order to be greater than $1/24$ of the LSB noise power. Capacitors of 100fF were initially used to be far above this requirement, and also to limit the effects of charge injection. This capacitance was later increased to maintain good stability and lower charge injection further. The settling period should be within half of the period (62.5 ns). Settling time was assumed to be 5 RC time constants; for $C=100\text{fF}$, the maximum switch resistance would be 125k . MOSFET switch sizings were therefore chosen based on the resistance of a NMOS and PMOS in parallel (transmission gate, shown in Figure 2).

Switch Design

Device sizes for the NMOS and PMOS were chosen based on resistance characterization (Figure 3). This test confirmed that by using a PMOS of $180\text{nm}/2\mu\text{m}$ and NMOS of $180\text{nm}/1\mu$ guarantees a low enough switch resistance of $3.03\text{k}\Omega$. When integrating the OTA into the switched capacitor, it was found that the capacitive loading of the feedback capacitors, combined with the resistance of the switch formed a pole and caused the amplifier to become unstable. To counteract this, a multiplier of 50 was applied to the switches active during the hold period to reduce this resistance, therefore raising the pole frequency by 50 times.

NMOS dummies were added to cancel the effects of charge injection. The size was swept to minimize the injection onto a 100fF capacitor. PMOS dummies were also initially tested with, but were found to not be required for the desired level of accuracy.

OTA Design

A 3 stage architecture was chosen for the OTA design. The multistage approach allowed us to keep device sizes small, enhancing bandwidth, while still allowing us to reach our target gain. It is worth noting that there is a natural tradeoff between the gain-bandwidth and the power consumption, and thus when designing our OTA, the goal was to meet our desired specifications with a margin of safety, but not exceed by

an unnecessary amount as to not increase power consumption. The three stages are a fully-differential 5-transistor OTA (M1-4), a twin pair of single-ended CS amplifiers (M6-7, M8-9), and a second fully-differential OTA (M10-14), as shown in Figure 4. The last stage contains Miller compensation, with a nulling zero to ensure stability. Overall the design has a gain of 73.7 dB, surpassing the gain target of 60dB and ensuring accuracy of the amplified signal.

The common mode feedback (CMFB, Figure 5) was designed after the main OTA. As the main OTA was designed around a bias voltage of $\sim 830\text{mV}$ in the nominal case (with the actual input being 900mV), the CMFB amplifier was designed to have 830 mV as the nominal output voltage. Near-minimum sizing was used for all the transistors in this design, to minimize the parasitic capacitances and increase bandwidth. This proved to be a stable design method, as the input pole of the CMFB ($100\text{ k}\Omega$ resistors in series) was far lower than any pole in the main OTA, giving a phase margin of 90 degrees. The 3dB bandwidth of the CMFB in open loop was 800 MHz , and while gain was not a priority in the CMFB amplifier design, the design achieved 25dB of DC open loop gain.

Results

The SC amplifier performed as expected, providing a double gain that is stable within a half-LSB margin of error. (This error limit ensures that the output data maintains perfect accuracy down to the desired bit resolution.) Figures 6-7 show the test benches for differential and CM signals. As indicated by the graph in Figure 8, the OTA is stable on differential operation (at gain crossover, the phase margin is 48.5 degrees). However, the low 3 dB bandwidth and nonlinear phase response (at frequencies greater than 10 kHz) means that high frequency signals will be delayed and attenuated. This could potentially cause signal distortion if the input is not a pure sine wave. Table 4 summarizes these results.

CMFB attenuates any CM variation by more than 17.5 dB (according to Figure 9), ensuring that the output CM level is stable at approximately 0.9 V . This is also verified by the full-scale and LSB step tests in the time domain (Figures 10-11).

The full system was tested using 4 input signals (Figure 12): a ramp across the entire operating range (+0.7 V to -0.7 V differential), a 2 MHz sine wave at full-scale (0.7 V) amplitude, and square waves of full-scale and LSB step size. In each case, the output voltage converges to the sampled values within half of a hold period. The zoomed plots (Figure 12B, D, F, H) confirm that the system is within a half-LSB error (5.5 mV), consistent with 8-bit accuracy.

Conclusion

An 8 MHz fully-differential switched-capacitor voltage doubler was successfully designed and implemented in TSMC 180 nm CMOS technology for use in 8-bit ADC applications. The system achieved the targeted 1.4 V differential output swing while maintaining operation from a 1.8 V supply and dissipating only 1.45 mW of power. The designed OTA achieved a DC gain of 73.7 dB, a gain crossover frequency of 321 MHz, and a phase margin of 48.5° , ensuring stable closed-loop operation. Time-domain simulations demonstrated that the amplifier settled within the required half-clock period and maintained errors below one-half LSB, satisfying the accuracy requirements for 8-bit conversion. The measured slew rate of 4.6 V/ns allowed the circuit to respond to full-scale input transitions in one clock cycle. System-level simulations verified proper operation for ramp, sinusoidal, full-scale step, and LSB step inputs. These prove the proper OTA behaviors, and the ability to interface the system with the given ADC specifications. Overall, this work demonstrates that a fully-differential SC voltage doubler can provide low-power, accurate signal amplification suitable for pipelined ADC applications in a 180 nm CMOS process.

We faced difficulties in the accuracy of the OTA during the hold period. The causes were determined to be charge injection, open loop gain, and stability. We found ringing in the signal, but we were able to improve the OTA to achieve the 1% accuracy and improve the stability. The three stage topology helped with increasing the gain and maintaining the bandwidth. We then implemented Miller compensation to achieve the final 48.5° phase margin. Further work could be done to improve the phase margin to at least 60° for better stability.

Figures and Tables

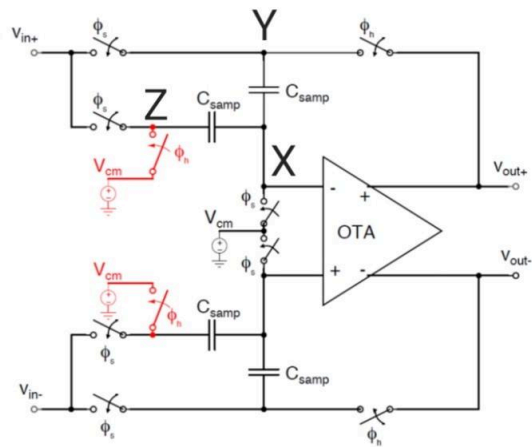


Figure 1. Circuit diagram for SC doubler.

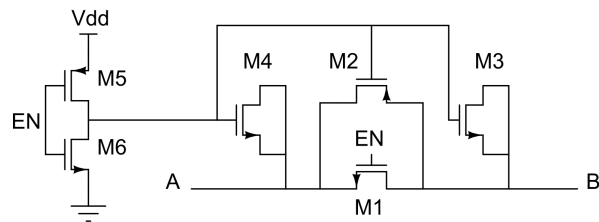


Figure 2. Schematic of CMOS switch.

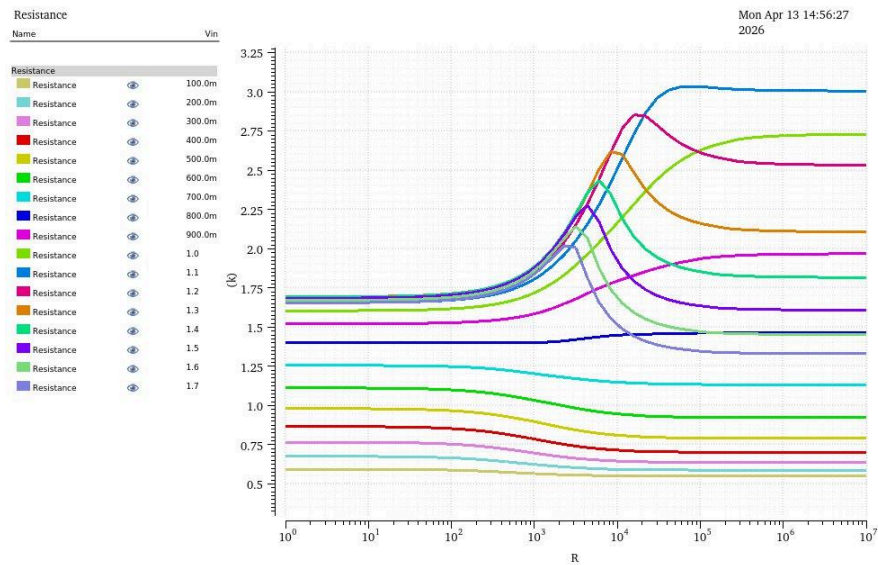


Figure 3. Resistance test for CMOS switch.

Table 1. CMFB OTA Design Values

Component Designator(s)	Width (m)	Length (m)
M1, M6	1u	180n
M2, M3, M4, M5	2u	180n

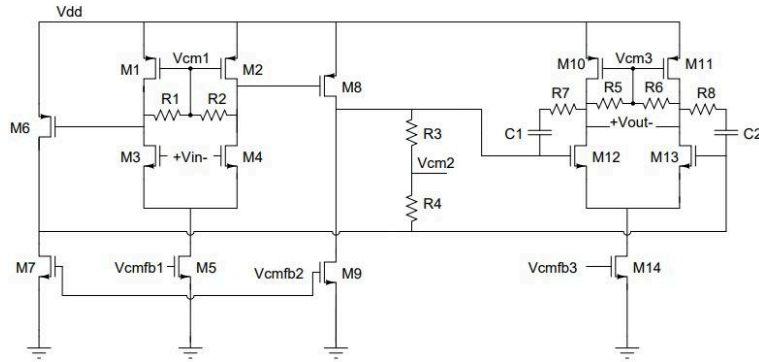


Figure 4. 3 Stage OTA Design

Table 2. Main OTA Component Values

Component Designator(s)	Width (μm)	Length (μm)	Value
M1-2 (stage 1 load)	15	0.5	—
M3-4 (stage 1 input)	8	0.5	—
M5,M14 (tail currents)	100	2	—
M6-9 (CS amplifiers)	1	0.5	—
M12-13 (stage 2 input)	7	0.5	—
M10-11 (stage 2 load)	14	0.5	—
R1,R2,R3,R4,R5,R6 (CM sensing)	—	—	100k Ω
R7,R8 (Miller compensation)	—	—	1.5k Ω
C1,C2 (Miller compensation)	—	—	400fF

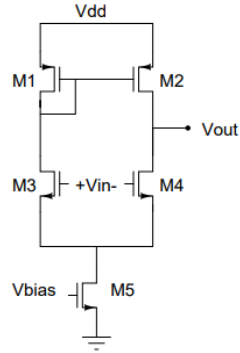


Figure 5. Common-Mode Feedback Amplifier Design

Table 3. CMFB Design Values

Component Designator(s)	Width (μm)	Length (μm)
M1,M2	0.22	0.18
M3,M4	0.3	0.18
M5	2	0.18

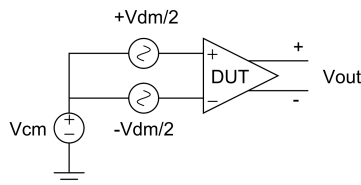


Figure 6. OTA Differential Mode Test Bench

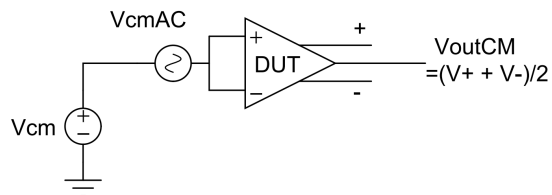


Figure 7. OTA Common Mode Test Bench

Table 4. Performance summary for OTA differential response.

Specification	Requirement	Result
DC Gain	> 60 dB	73.7 dB
3 dB Bandwidth		123 kHz
Gain Crossover	> 4 MHz (Nyquist BW)	321 MHz
Phase Margin		48.5 deg
Power		1.45mW
FOM		.71pJ/conversion step

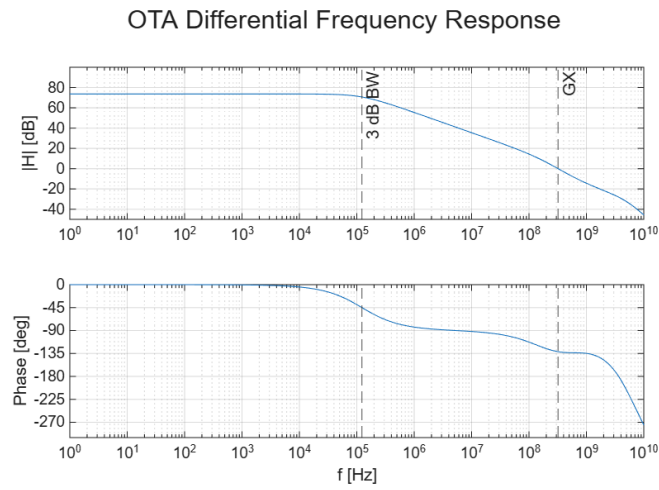


Figure 8. OTA differential-signal frequency response.

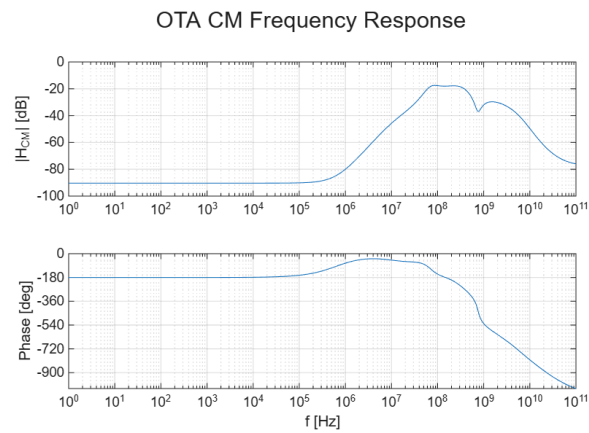


Figure 9. OTA CM frequency response, showing a maximum CM gain of -17.5 dB.

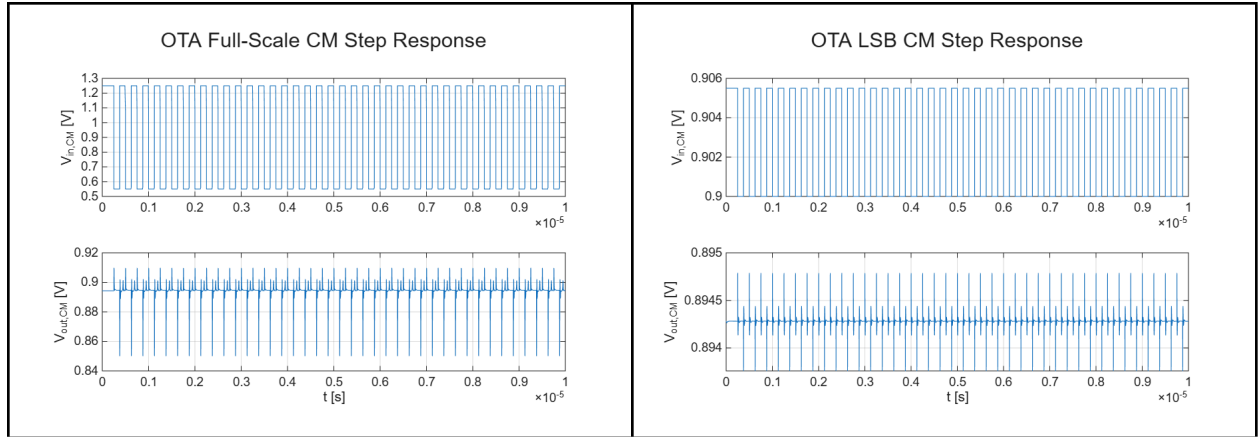


Figure 10. OTA step response for a full-scale input voltage (left) and LSB input (right).

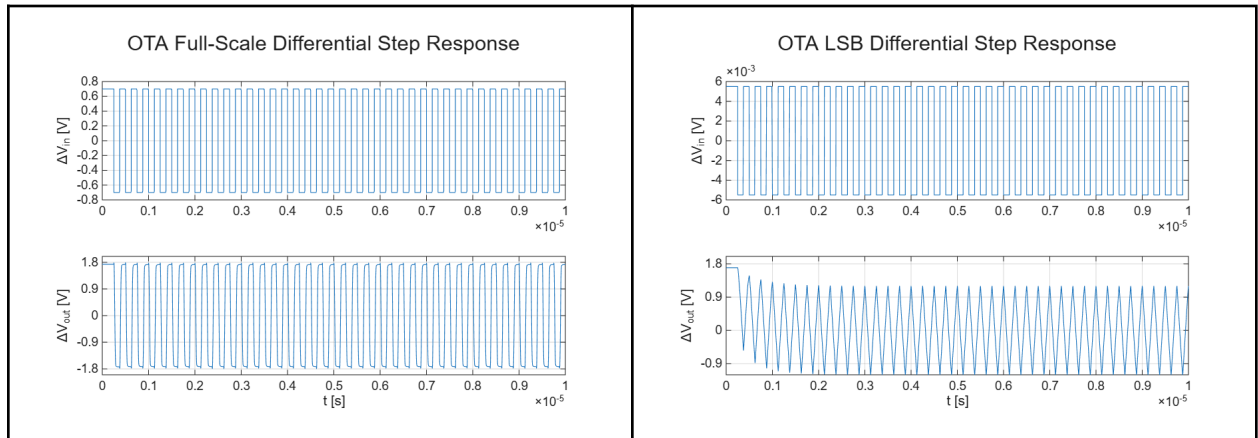
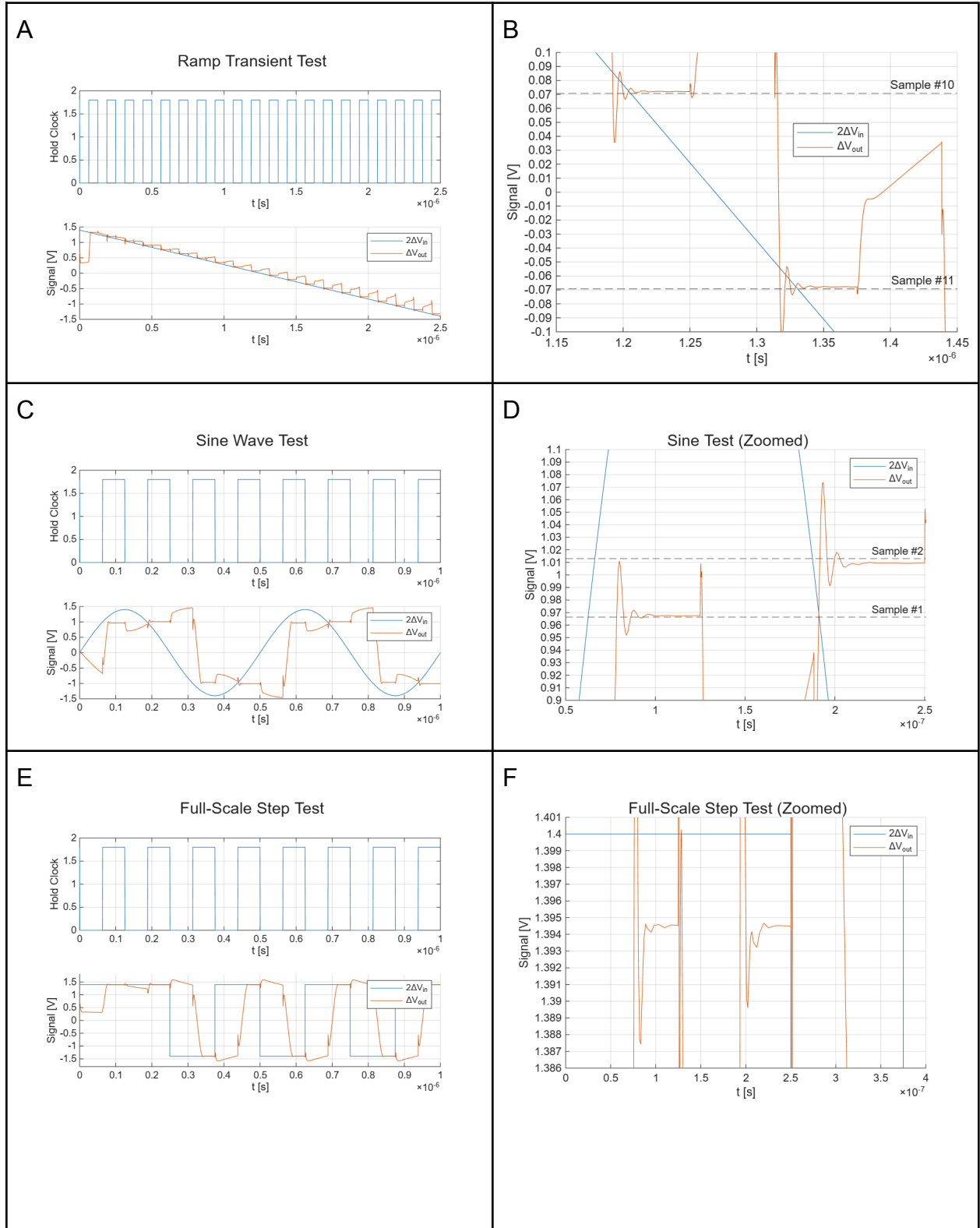


Figure 11. OTA differential step response for full-scale input voltage (left) and LSB input (right). The maximum slew rate of the OTA was measured from the full-step differential response to be 4.6 V/ns.

Figure 12. System test results for a full-range voltage ramp (A-B), 2 MHz sine wave (C-D), full-scale step (E-F), and LSB step (G-H).



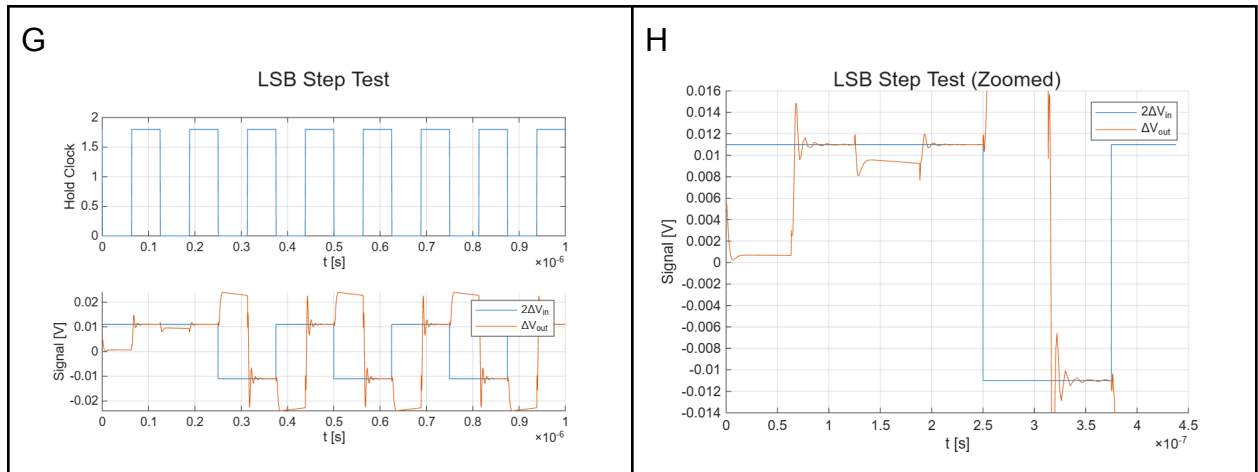


Figure 12 (cont.)